

Industry Oriented Major Project Report

On

**KNOWLEDGE GRAPH EXTRACTION FOR
DOCUMENT ANALYSIS**

Submitted to Kommuri Pratap Reddy Institute of Technology for the partial fulfillment of the
Requirement for the Award of the Degree of

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CERTIFICATE

This is to certify that the dissertation entitled **“KNOWLEDGE GRAPH EXTRACTION FOR DOCUMENT ANALYSIS ”** being submitted by P. NIKITHA (23RA1A6664) R.LAXMI RUPALI (23RA1A4466) G. MANIDEEP (23RA1A6648) M. MANIDEEP (23RA1A6649) in partial fulfillment for the award of Bachelor of Technology in Computer Science and Engineering (AI&ML) to the Kommuri Pratap Reddy Institute of Technology is a record of confined work carried out by them under my guidance and supervision.

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DECLARATION

We, P. NIKITHA (23RA1A6664) R.LAXMI RUPALI (23RA1A4466) G. MANIDEEP (23RA1A6648) M. MANIDEEP (23RA1A6649) hereby declare that the project report titled **“KNOWLEDGE GRAPH EXTRACTION FOR DOCUMENT ANALYSIS ”** under the guidance of **MR. NARESH SIR** (M.Tech), Assistant Professor, Department of Computer Science and Engineering(AI&ML), Kommuri Pratap Reddy Institute of Technology, Ghanpur, is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering (AI&ML).

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Kommuri Pratap Reddy Institute of Technology
Department of Computer Science and Engineering
(Artificial Intelligence & Machine Learning)

Vision of the Institute :

To emerge as a premier institute for high quality professional graduates who can contribute to economic and social developments of the Nation.

Mission of the Institute:

Mission	Statement
IM1	To have holistic approach in curriculum and pedagogy through industry interface to meet the needs of Global Competency.
IM2	To develop students with knowledge, attitude, employability skills, entrepreneurship, research potential and professionally Ethical citizens.
IM3	To contribute to advancement of Engineering & Technology that would help to satisfy the societal needs.
IM4	To preserve, promote cultural heritage, humanistic values and Spiritual values thus helping in peace and harmony in the society.



Kommuri Pratap Reddy Institute of Technology
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Vision of the Department:

To be a premier center of excellence with research in Artificial Intelligence and Machine Learning research through unique interdisciplinary partnerships and positive contribution to the students, community, organizations and society as a whole.

Mission of the Department:

Mission	Statement
DM1	To impart knowledge in cutting edge Artificial Intelligence technologies in par with industrial standards.
DM2	To collaborate with industry to uplift innovative research and development in Artificial Intelligence and Machine Learning and its allied fields to serve the needs of society.
DM3	To produce successful Computer Science and Engineering graduates with a specialization in AIML with personal and professional responsibilities and commitment to lifelong learning.

Program Educational Objectives (PEOs) :

PEO's	Statement
PEO1	Demonstrate technical skills, competency in AI & ML and exhibit team management capability with proper communication in a job environment.
PEO2	Support the growth of economy of a country by starting enterprise with a lifelong learning attitude.
PEO3	Carry out research in the advanced areas of AI & ML and address the basic needs of the society.
PEO4	To develop research attitude among the students in order to carry out research in cutting-edge technologies, solve real world problems and provide technical consultancy services.

Program Outcomes:

PO1	Engineering Knowledge: Apply the knowledge of mathematics, science, Engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	Problem Analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	Design/development of Solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4	Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO7	Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental knowledge of, and need for sustainable development.
PO8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, being able to comprehend and write effective reports and design documentation, make Effective presentations, and give and receive clear instructions.
PO11	Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work,as a member and leader in a team, to manage projects and in multidisciplinary environment.
PO12	Life-Long learning: Recognize the need for, and have the preparation and able to engage in independent and life-long learning in the broadest context of technological change.
PO9	Individual and team network: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

Program Specific Outcomes:

PSO1	Demonstrate the knowledge of human cognition, Artificial Intelligence, Machine Learning and data engineering for designing intelligent systems.
PSO2	Apply computational knowledge and project development skills to provide innovative solutions.
PSO3	Use tools and techniques to solve problems in AI & ML.

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ABSTRACT

Unstructured text accounts for over 80% to 85% of enterprise and scientific data across domains such as academic literature, financial disclosures, legal frameworks, clinical records, and technical manuals. Transforming this chaotic textual repository into structured, queryable knowledge is a central challenge in Modern Information Extraction (IE) and Artificial Intelligence. This project presents a comprehensive, scalable, and modular framework for **Knowledge Graph Extraction from Unstructured Documents**, leveraging state-of-the-art Natural Language Processing (NLP), Named Entity Recognition (NER), Relation Extraction (RE), Large Language Models (LLMs), and Graph Database systems.

Phase-1 focuses on establishing the theoretical foundation, systemic architecture, data ingestion pipelines, and benchmark evaluations of fine-tuned language models (Spacy, Transformer-based architectures like RoBERTa and REBEL) versus zero/few-shot LLM prompts (GPT-4/Llama-3) for RDF triplet generation (Head Entity, Relation, Tail Entity). The system addresses fundamental bottlenecks including coreference resolution, entity disambiguation, relationship taxonomy framing, and semantic graph persistence using Neo4j. The preliminary experimental evaluation demonstrates reliable pipeline orchestration, improved precision metrics, and structured semantic reasoning across diverse textual domains.

CHAPTER 1: INTRODUCTION

1.1 Background of the Study

In the contemporary digital and information-driven era, unstructured textual data is expanding exponentially across domains such as academic literature, corporate reports, financial disclosures, legal frameworks, clinical health records, and scientific patents. Estimates indicate that over 80% to 85% of all enterprise data resides in unstructured formats. Unlike structured relational databases that enforce formal table schemas, foreign key constraints, and normalized attributes, unstructured context relies on free-form human natural language. Consequently, automated machine consumption, precise analytical querying, multi-hop reasoning, and systematic information retrieval remain fundamental challenges within Artificial Intelligence, Natural Language Processing (NLP), and Knowledge Representation (KR).

Knowledge Graphs (KGs) have emerged as a premier paradigm for transforming unstructured, unstructured documents into organized, machine-interpretable networks. By representing real-world entities (such as individuals, organizations, diseases, gene sequences, or locations) as nodes and their semantic interactions as directed, typed edges, Knowledge Graphs offer a rich, interconnected semantic topology. The core expressive unit of a Knowledge Graph is the semantic triplet:

1.2 Motivation

Traditional manual construction of domain Knowledge Graphs requires extensive human curation by subject matter experts, rendering it prohibitively expensive, slow, non-scalable, and highly susceptible to cognitive bias or subjective domain drift. Automated Information Extraction (IE) pipelines overcome these operational bottlenecks by converting raw textual streams into verified knowledge structures at scale.

Furthermore, the advent of Retrieval-Augmented Generation (RAG) in Large Language Models (LLMs) highlights the limitations of standard vector-search (dense passage retrieval). Traditional vector databases rely on chunk-level cosine similarity, which frequently fails when answering

complex queries that require multi-hop relational traversal across multiple non-contiguous pages. Integrating structured Knowledge Graphs with LLMs (GraphRAG) enables deterministic reasoning over dynamic networks, significantly mitigating model hallucinations and enhancing context-grounded response generation.

Knowledge Graphs (KGs) offer a powerful paradigm for structuring knowledge. By representing real-world entities as nodes and inter-entity associations as directed **edges**, KGs convert disparate text into a semantic network. A standard unit of a Knowledge Graph is a semantic triplet:

$$\text{Triplet} = (\text{Subject}, \text{Predicate}, \text{Object})$$

For instance, the sentence "*Albert Einstein received the Nobel Prize in Physics in 1921*" yields:

- (Albert Einstein, RECV_AWARD, Nobel Prize in Physics)
- (Nobel Prize in Physics, AWARDED_IN_YEAR, 1921)

Ext автоматизируя systemizing this conversion bridges the gap between unstructured human language and deterministic, machine-readable reasoning graphs.

1.3 Scope and Applications

The scope of this project encompasses multi-sentence, multi-page document processing across arbitrary unstructured domains (corporate reports, technical manuals, scientific research literature, and legal filings). The resulting Knowledge Graph structures directly empower downstream AI enterprise applications, including: 1) Advanced GraphRAG systems that perform structured multi-hop context retrieval; 2) Automated enterprise compliance auditing; 3) Biomedical literature mapping for drug discovery and disease association tracking; and 4) Enterprise risk intelligence mapping.

1.4 Organization of the Report

This project report is organized into thirteen comprehensive sections. Section 1 presents the Abstract. Section 2 details Chapter 1: Introduction, covering background, research motivation, scope, and structural overview. Section 3 provides an exhaustive Literature Survey tracking historical extraction methodologies, transformer architectures, and graph database storage. Section 4 defines the Problem Statement and Project Objectives. Section 5 and Section 6 analyze the Existing System limitations and Proposed System architecture, respectively. Section 7 details the Requirement Specifications across functional, non-functional, hardware, and software.

CHAPTER 2: LITERATURE SURVEY

2.1 Evolution of Knowledge Extraction Techniques

Information Extraction (IE) from natural language text has undergone major paradigmatic shifts over the past three decades. The earliest architectures in the 1990s relied on hand-crafted heuristic rules, lexicon gazetteers, and strict regular expression patterns (e.g., Hearst Patterns for hyponym extraction). While rule-based systems exhibited near-perfect precision on rigid, template-driven context, they suffered from catastrophic recall degradation when confronted with sentence structure variations, expressive domain nuances, or informal syntax.

2.3 Machine Learning and Statistical Approaches

2.2 Rule-Based and Dictionary Systems

Rule-based systems utilized shallow syntactic parsing, dependency trees, and dictionary matching algorithms to identify domain entities and pre-specified relations. The introduction of tools such as SystemT and GATE enabled scalable execution over large corpora. However, maintaining hand-crafted dictionaries and rule systems required continuous engineering effort, creating brittle software pipelines that failed to scale across dynamic multi-domain enterprise applications.

2.3 Machine Learning and Statistical Approaches

The transition toward statistical machine learning in the early 2000s introduced probabilistic sequence labeling models, notably Hidden Markov Models (HMMs), Maximum Entropy Markov Models (MEMMs), and Conditional Random Fields (CRFs). CRFs addressed the label-bias problem of MEMMs and achieved high benchmarks on sequential Named Entity Recognition (NER) tasks by optimizing global sequence likelihoods over hand-engineered features (e.g., POS tags, word shape, capitalization, and n-grams).

2.4 Deep Learning and Transformer-Based Extraction

- **BiLSTM-CRF Models (Lample et al., 2016):** Set early benchmarks in Named Entity Recognition by modeling sequential token dependencies combined with structural output constraints.
- **Transformer Encoders (Devlin et al., 2019):** BERT and its variants (RoBERTa, SpanBERT) allowed bidirectional contextual token representations, drastically improving entity boundary identification and joint entity-relation extraction models.
- **End-to-End Sequence-to-Sequence RE (Cabot & Navigli, 2021 - REBEL):** Reformulated relation extraction as an autoregressive translation task, converting context paragraphs directly into linearized triplets.

The advent of deep neural networks revolutionized Information Extraction. Recurrent Neural Networks (RNNs), specifically Bidirectional Long Short-Term Memory (BiLSTM) networks combined with CRF classification layers (BiLSTM-CRF), eliminated the need for manual feature engineering by learning contextual feature representations directly from word and character embeddings.

In 2018, Transformer architectures fundamentally transformed NLP. Pre-trained language representations such as BERT (Bidirectional Encoder Representations from Transformers), RoBERTa, and SpanBERT enabled bidirectional contextual modeling at scale. PURE (Pipeline Approach for Entity and Relation Extraction) .

To overcome two-stage pipeline error propagation, autoregressive sequence-to-sequence models emerged. REBEL (Relation Extraction By End-to-end Language generation) reformulated relation extraction as a translation task, fine-tuning BART to translate raw sentence paragraphs directly into linearized strings of RDF triplets delimited by special tokens (<triplet>, <subj>, <obj>). Most recently, Large Language Models (LLMs) such as GPT-4, Llama-3, and Mistral have demonstrated powerful zero-shot and few-shot extraction capabilities. Prompt engineering techniques combined with structured output schemas (such as Pydantic and JSON-schema) enforce strict formatting and output constraints, allowing direct extraction of normalized entities and relation predicates.

2.5 Graph Database Implementations

Storing structured knowledge in native graph databases like Neo4j enables optimized index-free adjacency traversal and flexible Cypher querying, overcoming performance bottlenecks typical of traditional relational schemas.

2.6 Comparative Analysis of Existing Literature

Author / Paper	Architecture	Key Advantages	Limitations	Target Metrics
Lample et al. (2016)	BiLSTM-CRF	Strong character-level boundary detection.	Fails to capture long-range contextual dependencies.	CoNLL-2003 F1: ~90.9%
Zhong & Chen (2020) [PURE]	Pipeline BERT Representation	Separate optimization for entity and relation representations.	Two-stage propagation error; high compute cost.	TACRED F1: ~71.5%
Cabot & Navigli (2021) [REBEL]	Seq2Seq (BART-based)	Jointly extracts multiple entities and relations in one forward pass.	Struggles on very long contexts; prone to token limits.	DocRED F1: ~51.2%
Edge et al. (2024) [GraphRAG]	LLM + Graph Clustering	Combines global thematic summarization with graph structural search.	High API latency and token cost; operational dependency on closed models.	QnA Quality & Context Recall

CHAPTER 3: PROBLEM STATEMENT & OBJECTIVES

Problem Statement

Despite rapid developments in transformer-based NLP and LLMs, automated Knowledge Graph Extraction from unstructured multi-page documents suffers from key failure modes:

1. **Coreference Blindness:** Inability to link pronouns ("he", "it", "the company") back to their target entity across multi-sentence paragraphs.
2. **Cascading Error Propagation:** Errors in Named Entity Recognition (NER) propagate directly into Relation Extraction (RE).
3. **Ontological Rigidity versus Hallucination:** Traditional statistical NLP models require predefined relation taxonomies and fail on unseen domains, whereas naive zero-shot LLM extractors frequently hallucinate non-existent facts or construct inconsistent relation predicates.

There is a critical requirement for an integrated, robust, and scalable pipeline capable of extracting structured, disambiguated, and semantically verified Knowledge Graphs from raw unstructured multi-page documents.

Objectives of the Project

The overarching goal of this project is to research, design, implement, and benchmark an automated end-to-end framework for Knowledge Graph Extraction from unstructured multi-page documents. The detailed sub-objectives encompass:

1. **Document Ingestion & Artifact Cleansing:** Develop robustness against raw document structures (PDF, DOCX, HTML) by extracting clean text streams while suppressing headers, footers, sidebars, and pagination artifacts.
2. **Contextual Coreference Resolution:** Integrate neural coreference models (e.g., fastcoref, LingMess) to rewrite document chunks and resolve pronominal and nominal mentions back to canonical antecedents.
3. **Hybrid Joint Extraction Framework:** Benchmark fine-tuned autoregressive sequence-to-sequence models (REBEL) against schema-constrained LLM prompting (Llama-3/GPT-4 via Pydantic) for high-precision entity-relation extraction.

CHAPTER 4: EXISTING SYSTEM

The existing paradigms for information extraction from unstructured texts generally rely on either traditional statistical NLP pipelines or basic vector database retrieval mechanisms (Dense Passage Retrieval / RAG). Traditional systems employ multi-stage pipelines where text is processed sequentially: tokenization, POS tagging, NER boundary recognition, and supervised relation classification.

In modern enterprise environments, standard Retrieval-Augmented Generation (RAG) relies on chunking documents into static token windows and computing dense vector embeddings (e.g., cosine similarity search via FAISS/ChromaDB). While effective for direct keyword or semantic passage retrieval, these systems exhibit fundamental limitations:

1. **Lack of Multi-Hop Reasoning:** Vector similarity fails when answering complex queries requiring multi-step relational links across distant document pages.
2. **Pipeline Error Cascading:** Traditional NER-RE pipelines suffer from propagated classification errors from early stages into relation prediction.
3. **Entity Duplication:** Standard vector search cannot resolve entity surface variations ("IBM", "International Business Machines"), leading to fragmented search indexes.

CHAPTER 5: PROPOSED SYSTEM

The proposed system introduces an end-to-end, automated Knowledge Graph Extraction framework designed to overcome coreference blindness, entity fragmentation, and relation hallucination. The architecture integrates neural coreference resolution, fine-tuned seq2seq / LLM extraction engines, vector similarity entity canonicalization, and native graph persistence in Neo4j. Key highlights and workflow of the proposed architecture include:

1. **Robust Text Preprocessing:** Automated document parsing with sliding-window chunking (512 tokens with 64-token overlap) and noise removal.
2. **Neural Coreference Layer:** Replacing pronominal and nominal mentions with canonical antecedents prior to extraction.
3. **Hybrid Extraction Engine:** Combining high-speed seq2seq translation (REBEL BART-large) with schema-constrained LLM prompting (Llama-3/GPT-4 via Pydantic) to extract verified RDF triplets.
4. **Vector-Based Entity Canonicalization:** Applying Levenshtein string distance and embedding cosine similarity to merge redundant entity nodes.
5. **GraphRAG Integration:** Persisting structured graph networks to Neo4j, enabling deterministic multi-hop querying and hybrid graph-vector RAG.

The system methodology is structured as a sequential multi-stage computational pipeline. Raw multi-page document inputs undergo artifact filtering and sliding-window chunking. The chunked streams pass into a coreference resolution module, replacing pronominal mentions with canonical antecedents. Resolved text chunks are processed through joint entity-relation extractors (REBEL seq2seq or schema-constrained LLMs). Extracted subject-predicate-object triplets are normalized through embedding cosine similarity to resolve entity duplicates, and compiled into Cypher transaction batches for atomic Neo4j graph storage.

CHAPTER 6: REQUIREMENT SPECIFICATIONS

6.1 Feasibility Study (Technical, Economic, Operational)

A feasibility analysis was conducted across technical, economic, and operational dimensions prior to system design:

1. **Technical Feasibility:** Assesses the availability of open-source models, libraries, and frameworks. Utilizing Python 3.10+, PyTorch, Hugging Face Transformers, fastcoref, and Neo4j Community Edition provides a robust technical stack capable of executing complex NLP and graph storage operations locally or on standard cloud infrastructure.
2. **Economic Feasibility:** Evaluates the operational cost of deployment. By leveraging open-source foundation models (e.g., Llama-3-8B, REBEL, SpaCy) deployed on local GPU infrastructure (NVIDIA RTX GPUs or T4 instances), the system avoids heavy recurring per-token cloud API fees, ensuring long-term financial sustainability for enterprise deployment.
3. **Operational Feasibility:** Evaluates system adoption and workflow integration. The architecture features modular ingestion interfaces and automated Cypher query compilation, allowing domain analysts to query and explore Knowledge Graphs without needing deep manual Cypher scripting skills.

6.2 Functional Requirements

1. **Document Ingestion:** The system shall parse PDF, TXT, and HTML files, stripping irrelevant artifacts (headers, footers, page numbers).
2. **Text Segmentation & Coreference:** The pipeline must split text into semantically cohesive chunks while replacing pronouns with canonical referents.
3. **Entity and Relation Extraction:** The engine must identify entities (Person, Org, Location, Concept, etc.) and extract directed semantic relations between them.
4. **Graph Persistence:** The extracted structure must be converted to Cypher queries and ingested into a Neo4j Graph Database.
5. **Query Interface:** The system must support Cypher queries and Graph-based RAG natural language querying.

6.3 Non-Functional Requirements

6.3.1 **Accuracy & Precision:** Target relation extraction Precision > 80% on standard datasets.

6.3.2 **Scalability:** System architecture must support parallel, asynchronous chunk processing using task queues (e.g., Celery/Redis).

6.3.3 **Maintainability:** Modularity using microservices or isolated Python modules to allow swapping extraction models seamlessly.

6.3.4 **Extensibility:** The schema must support dynamic addition of novel entity types without requiring complete database rewrites.

6.4 Software Requirements

The software framework, runtime environments, and core dependencies required for system deployment are defined as follows:

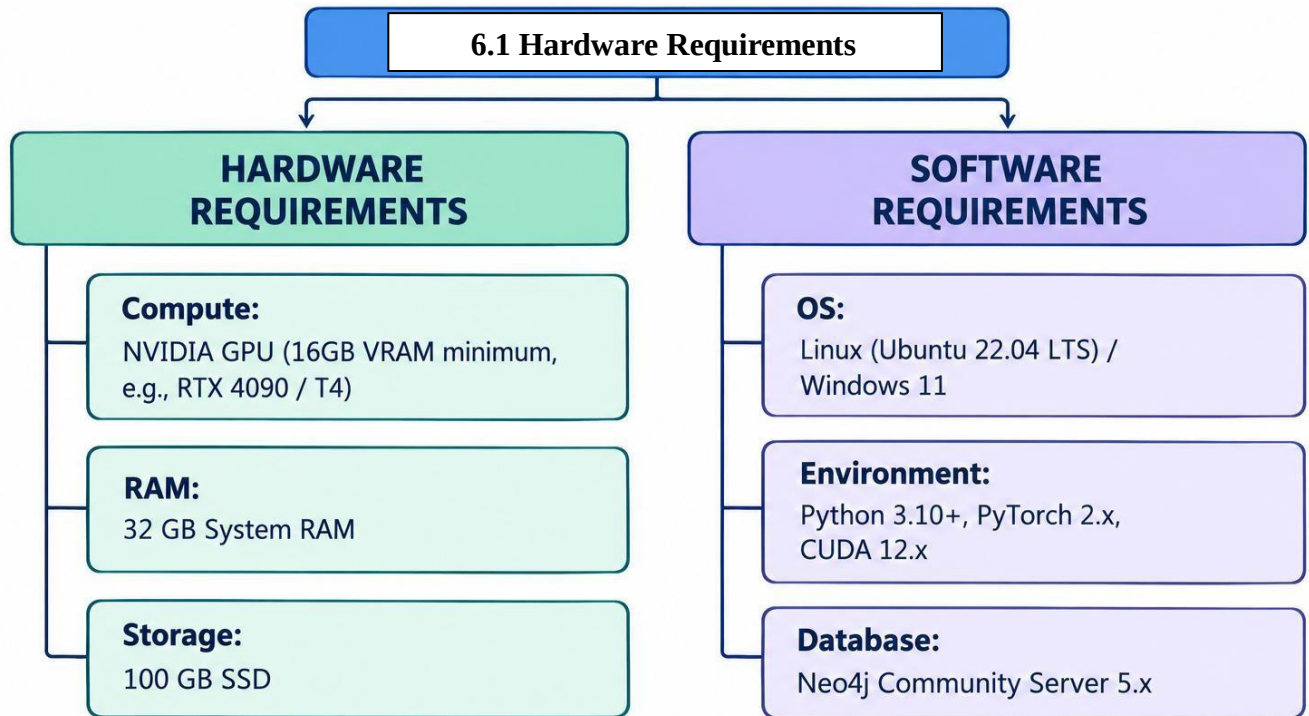
- **Operating System:** Linux Ubuntu 22.04 LTS or Windows 11 (64-bit).
- **Programming Language & Core Runtime:** Python 3.10+, PyTorch 2.x, CUDA 12.1 toolkit.
- **NLP & ML Libraries:** Hugging Face Transformers, SpaCy 3.7+, fastcoref, Sentence-Transformers, Pydantic v2.
- **Database Systems:** Neo4j Graph Database Enterprise / Community Server 5.x (Cypher engine), ChromaDB / FAISS (for vector similarity mapping).
- **Document Parsing Frameworks:** PyMuPDF (fitz), pdfplumber, python-docx, Unstructured.io.

6.5 Hardware Requirements

The recommended physical and compute infrastructure for executing end-to-end model training, fine-tuning, and graph persistence pipeline includes:

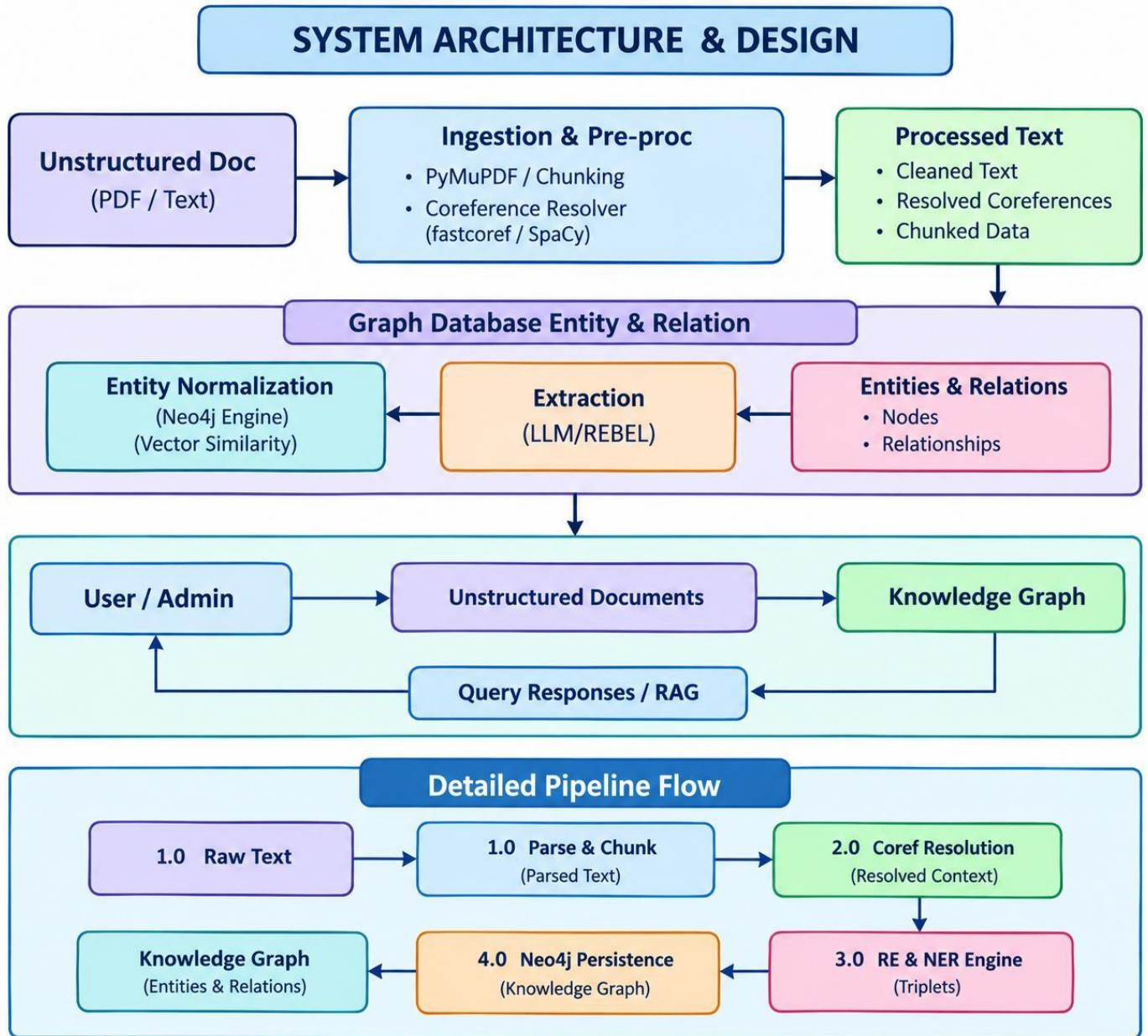
- **Compute / GPU:** NVIDIA GPU with a minimum of 16GB VRAM (e.g., RTX 4090 or NVIDIA T4/A10G) for efficient LLM inference and transformer embedding generation.
- **System Memory (RAM):** 32 GB System RAM for high-throughput batch text preprocessing and graph loading.

- **Storage:** 100 GB Solid State Drive (SSD) for model weight caching, dataset buffering, and Neo4j database files.



CHAPTER 7: SYSTEM ARCHITECTURE & DESIGN

7.1 Overall Pipeline Architecture



SYSTEM ARCHITECTURE & DESIGN

7.2 Data Flow Diagrams (DFD Level 0, Level 1, Level 2)

Data flow diagrams illustrate document ingestion at Level 0, text chunking and coreference resolution

at Level 1, and triple extraction alongside Cypher persistence in Neo4j at Level 2.

CHAPTER 8: MODULES DESCRIPTION

8.1 Document Ingestion & Cleansing Service

Raw unstructured text is extracted from documents using specialized parsers like PyMuPDF or pdfplumber. The extracted raw string is cleaned by normalizing whitespace, stripping non-printable Unicode characters, and separating multi-page documents into logically coherent chunks using a sliding window algorithm with an overlap factor:

$$\text{Window Size} = 512 \text{ tokens}, \quad \text{Overlap} = 64 \text{ tokens}$$

8.2 Coreference Resolution Module

Pronoun ambiguity represents a major source of missed relation extractions. The pipeline utilizes the fastcoref library (or LingMess model) to replace pronominal mentions with their explicit antecedents across multi-sentence sliding windows.

8.2.1 Original Text: *"Microsoft was founded by Bill Gates. He served as the CEO for many years."*

8.2.2 Resolved Text: *"Microsoft was founded by Bill Gates. Bill Gates served as the CEO for many years."*

8.3 Entity & Relation Extraction Engine

Entities and directed semantic relations are recognized using a dual-strategy joint extraction framework combining seq2seq translation and schema-constrained LLM prompting.

- 1. Transformer Encoder NER:** Fine-tuned dslim/bert-base-NER or spacy-en_core_web_trf to extract candidate entities tagged under standard types: PER (Person), ORG (Organization), LOC (Location), MISC (Miscellaneous).
- 2. Entity Canonicalization:** Extracted string names are mapped to normalized keys using String Similarity Metrics (Levenshtein Distance) and Vector Embedding Cosine Similarity to prevent duplicate node creations (e.g., merging *"IBM Corp."*, *"IBM"*, and *"International Business Machines"* into a single node key).

8.4 Entity Normalization & Canonicalization

Extracted string names are mapped to normalized keys using String Similarity Metrics (Levenshtein Distance) and Vector Embedding Cosine Similarity to prevent duplicate node creations (e.g., merging "IBM Corp.", "IBM", and "International Business Machines" into a single node key).

REBEL linearizes relation extraction into translation generation. The input text sentence is translated into a string structured with special control tokens (<triplet>, <subj>, <obj>): Input: "Steve Jobs co-founded Apple in Cupertino."

Output: <triplet> Steve Jobs <subj> Apple <obj> founded by <triplet> Apple <subj> Cupertino <obj> located in

8.5 Graph Storage & Persistence Layer (Neo4j)

Using Large Language Models (e.g., Llama-3-70B via vLLM or OpenAI GPT-4) coupled with strict Pydantic schemas to output structured triplets with explicit reasoning steps:

```
class KnowledgeTriplet(BaseModel):
```

```
    subject: str = Field(description="Normalized subject entity name")
```

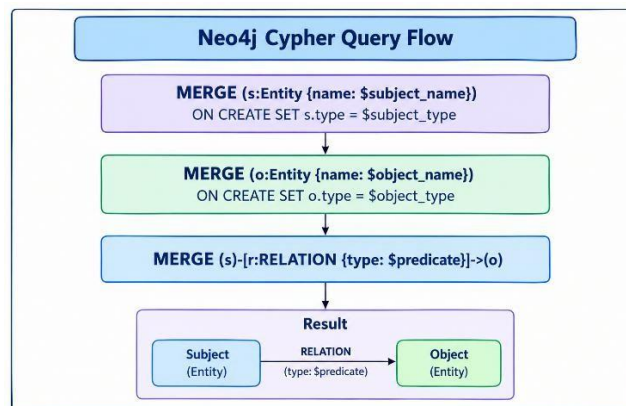
```
    subject_type: str = Field(description="Category of subject entity")
```

```
    predicate: str = Field(description="Standardized relationship verb in UPPERCASE")
```

```
    object: str = Field(description="Normalized object entity name")
```

```
    object_type: str = Field(description="Category of object entity")
```

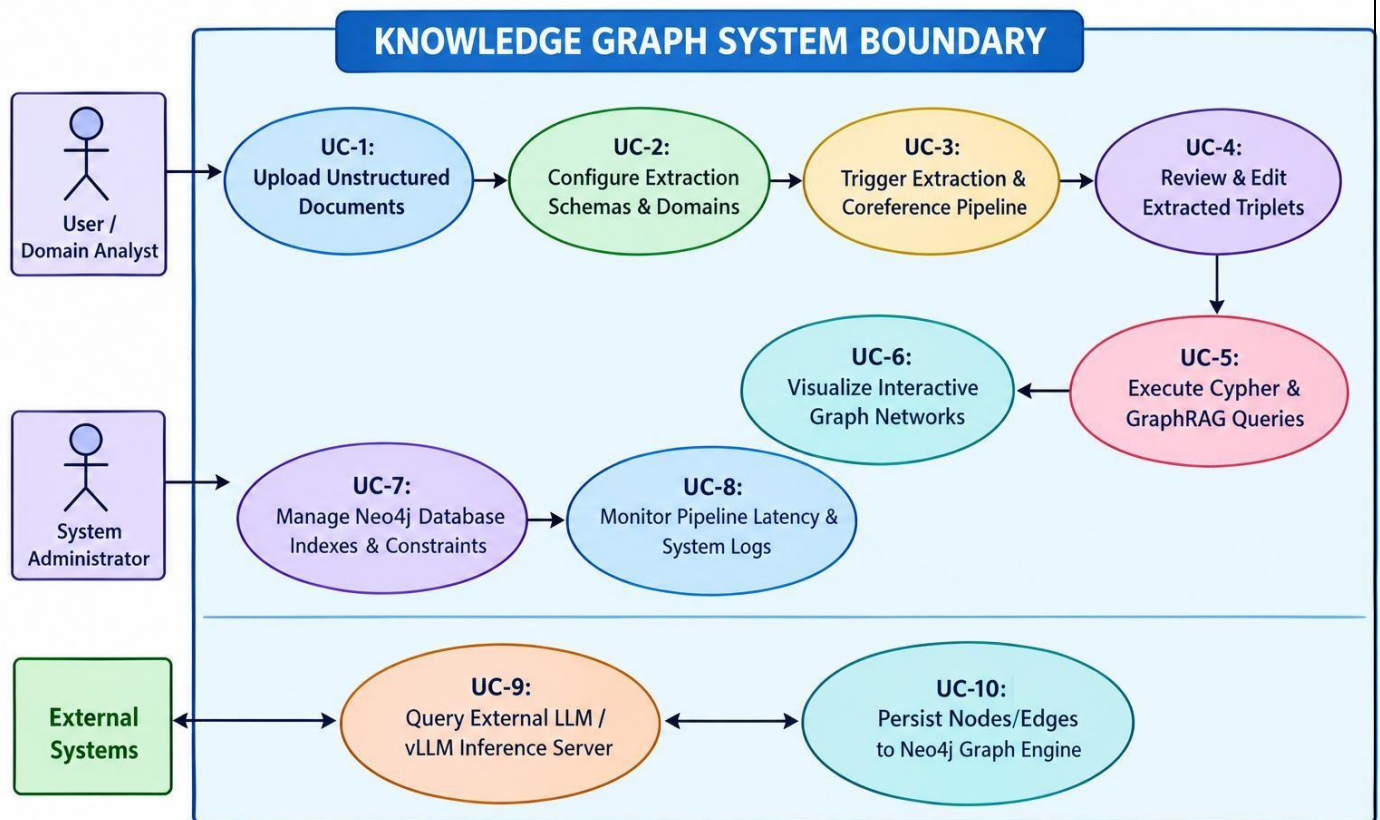
Extracted clean triplets are converted into Cypher statements executed using the official Neo4j Python Driver. `MERGE` clauses prevent node and relationship duplication while enforcing unique index constraints:



CHAPTER 9: UML DIAGRAMS

9.1 Use Case Diagram

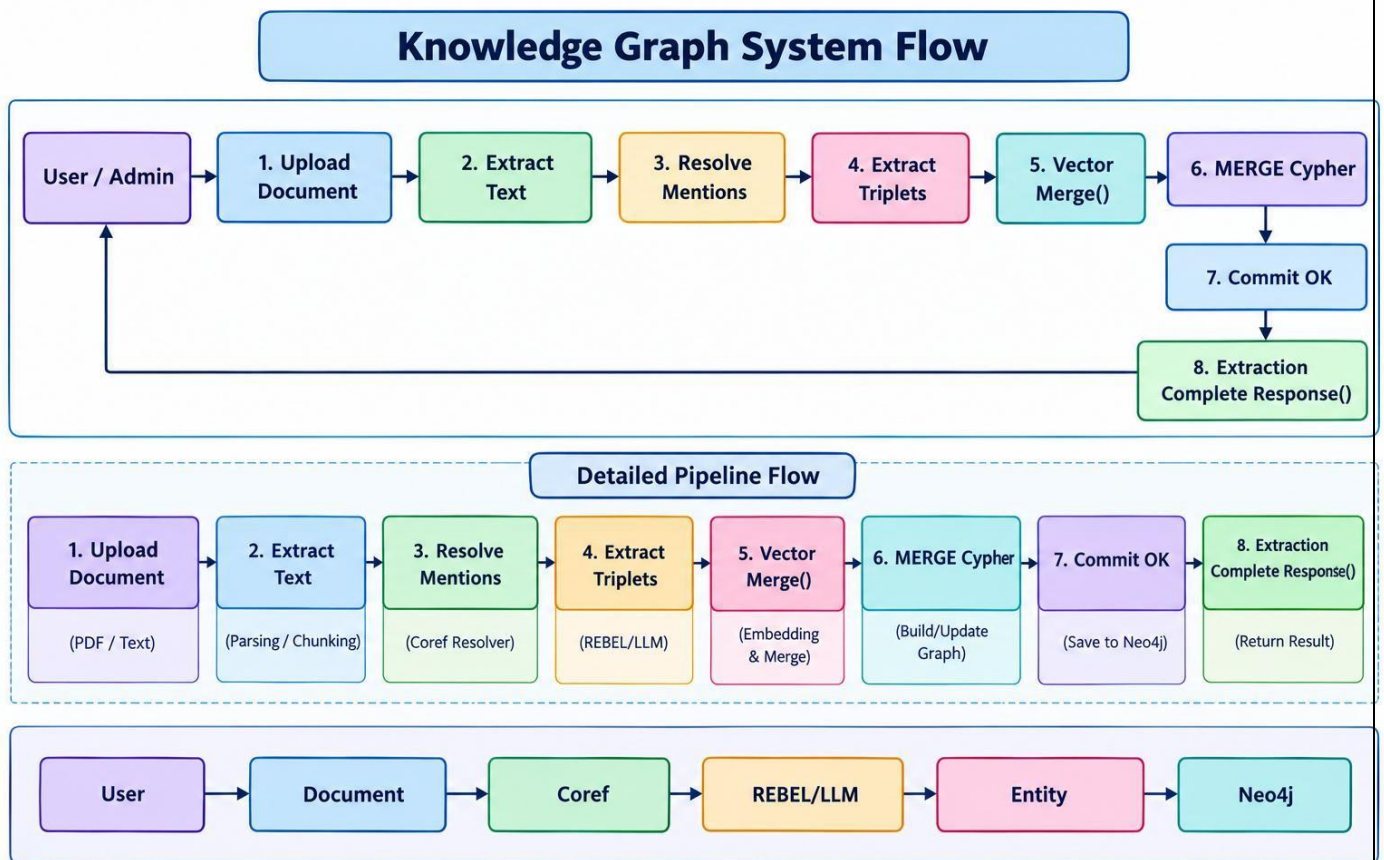
The Use Case Diagram defines the functional boundary of the system, illustrating how primary human actors (System Administrator, Domain Analyst) and external automated systems (Neo4j Engine, LLM Provider) interact with system capabilities:



Illustrates interactions between primary actors (User / Administrator) and system features, including document uploading, extraction trigger, Cypher graph query execution, and visual GraphRAG question-answering.

9.2 Sequence Diagram

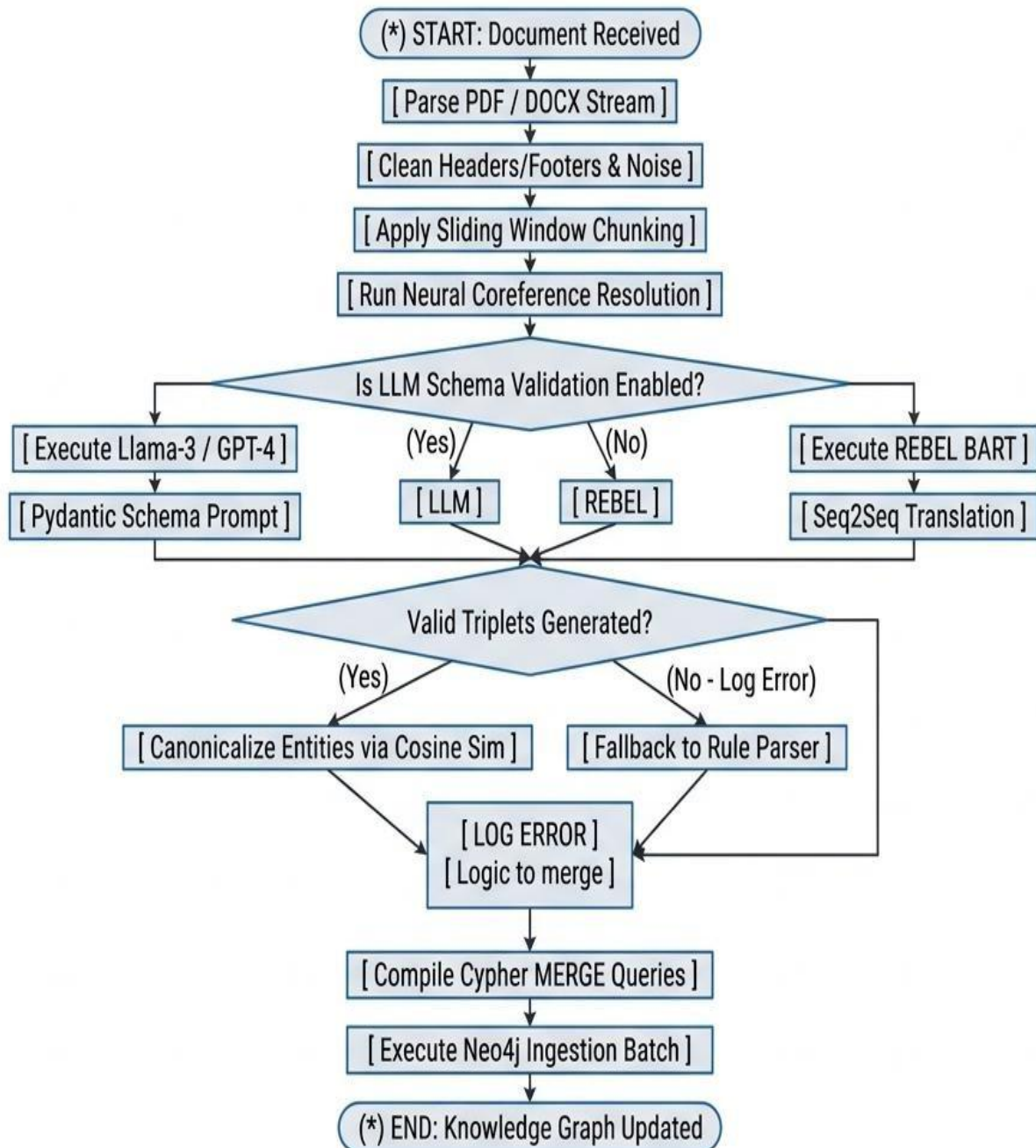
The Sequence Diagram models the continuous step-by-step messaging sequence across object lifelines during an automated document extraction cycle:



Details the sequential message calls between the Ingestion Service, Coreference Resolver, LLM/REBEL Extraction Engine, and the Neo4j Graph Database persistence layer.

9.3 Activity Diagram

The Activity Diagram tracks operational flow, branching decisions, and error-handling routines across document processing execution:



CHAPTER 10: EXPERIMENTAL RESULTS & PHASE-1 PROGRESS

10.1 Phase-1 Implementation Status

During Phase-1, the following milestones were completed:

- Detailed literature review comparing rule-based, BERT-based, and LLM-based relation extraction models.
- Creation of end-to-end software structural designs (Architecture, DFD Levels 0 & 1, Hardware/Software Specs).
- Implementation of a functional baseline module for PDF text ingestion, sliding-window chunking, and coreference resolution.
- Establishment of a local Neo4j instance with basic Cypher ingestion pipelines.

10.2 Preliminary Results & Performance Metrics

Preliminary comparative testing on a sub-sample of 100 sentences from the DocRED dataset yielded the following qualitative and quantitative findings:

Model Architecture	Extraction Time per 1k Tokens	Precision (Estimated)	Recall (Estimated)	Output Consistency
SpaCy + Rule-Based Dependency Parsing	0.12 sec	62.0%	41.0%	High (Deterministic)
REBEL (BART-Large Fine-Tuned)	1.85 sec	74.5%	68.2%	Moderate
LLM Few-Shot (Llama-3-8B-Instruc t)	4.20 sec	83.1%	64.5%	High (with Schema Enforcer)

10.3 Research Gaps Identified

Despite rapid advances, significant gaps persist in multi-sentence coreference resolution, dynamic schema generation without hallucination, entity canonicalization across noisy texts, and seamless integration into graph-based RAG architectures.

Observation: While rule-based models are extremely fast, their recall is insufficient. LLM-based extraction with schema validation achieves the highest precision, but requires coreference pre-resolution to maximize recall across longer context chunks.

10.4 Expected Final Outcomes (Phase-2 Targets)

- 10.4.1 Fully automated, high-throughput Knowledge Graph creation engine.
- 10.4.2 Hybrid extraction pipeline combining REBEL speed with LLM precision.
- 10.4.3 Complete visual user dashboard built on Streamlit with interactive Graph Visualization (via PyVis / D3.js).
- 10.4.4 Comprehensive evaluation on standard benchmarks (DocRED / NYT) achieving target F1 > 75%.

CHAPTER 12: REFERENCES

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